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Review article

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Wearable haptics for virtual reality and beyond

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Abstract

Research has increasingly focused on wearable systems that provide informative haptic sensations to the hands, enhancing user interaction in virtual reality (VR) and augmented reality (AR) environments. Concurrently, the evolution of VR visors in terms of wearability, quality of the visual immersion and immediateness of bare-hand interaction poses both opportunities and challenges for haptic systems, which must match these visors in usability, hand-tracking compliance and quality of experience. Designing haptic devices involves trade-offs between the dimensions and quality of feedback, often falling short of the richness and complexity of natural haptic sensations. Recent studies have explored various actuating methods and haptic prototypes to enhance the wearability and feedback richness of haptic devices. Notably, the development of soft interfaces actuated by diverse principles has emerged as a key trend. Moreover, beyond simulating physical interaction, haptic technologies can provide high-level information for various applications, including posture correction, navigation and remote control of humanoid robots. In this Review, we provide an overview of wearable haptics for VR applications, introducing principles of haptic rendering and the relationship between stimulus modalities and actuating solutions. We discuss emerging actuating methods and wearable device designs, examining their experimental applications in innovative scenarios of VR and beyond.

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Key points

- Various haptic stimuli and mechanoreceptors are involved in the sense of touch and in natural interaction with physical objects. These can be mapped to different actuation modalities adopted by wearable device designs.
- Haptic rendering in immersive virtual reality (VR) is a trade-off between the plurality of haptic stimuli and the constraint provided by physical wearable devices. Certain rendering principles can take advantage of those haptic features that humans are most sensitive to.
- According to actuation principles and target applications, a plurality of haptic designs is proposed in the literature, covering different actuation methods and different body segments, from the finger pad to the limbs.
- Although rendering in VR is a target application for wearable device design, innovative applications are proposed in several other fields, ranging from postural correction and navigation assistance to communication and remote control of humanoid robots.
- Research in wearable haptics is rich in innovative actuators, devices and applications, but certain fundamental haptic features, such as the stick–slip condition and wearability of devices compared with rendered features, still represent an unmet need and a challenge to foster the use of wearable haptics in VR and in other innovative applications.

Introduction

The evolution of virtual reality (VR) with the notion of extended reality (XR), including the metaverse, has substantially amplified interest towards wearable devices that can offer the users tactile sensations directly, and that can be seamlessly integrated with the human body, thereby enhancing the immersive experience in XR environments. VR involves immersion in a purely virtual world, usually with interaction mediated by handheld controllers. Since about 2010, the evolution of hand-tracking technologies has been substituting such a mediated interface with direct, bare-hand interaction. XR merges VR and augmented reality (AR), allowing for the manipulation of physical objects, virtual objects and augmented physical objects within the same scenario.

With haptics, augmentation extends beyond visual information to include tactile information, fundamentally transforming interactions between users and objects. This experience has been enabled by high-performance hand-tracking technologies that require haptics devices to be designed to cover or to minimally interfere with the finger pad through thin, pliable actuated layers. Recent innovations in design solutions have enabled the development of minimalistic vibrotactile devices. These devices resemble the form factor of earpods, seamlessly adhere to the dorsum of the fingertip and house embedded electronics. This paradigm shift aims to heighten dexterity and manipulative capabilities, fostering an almost hands-free experience within the XR landscape. Additionally, such design choices must find compromises with the variety and complexity of the concurrent mechanical stimuli (that is, contact perception resembled by normal skin deformation only, hence not including lateral stretch, surface orientation, edges and similar spatially distributed cues), naturally perceived through the sense of touch, and with the intrinsic limitations that wearable devices impose, in terms of mechanical constraints and feasible rendering features. Beyond simulation of the tactile perception in virtual interactions, growing interest has emerged in applications exploiting the sense of touch to convey high-level information to the user, hence freeing other sensory pathways involved in a given task (Fig. 1). These applications usually change the interaction point from hands to other body segments, in order not to limit the dexterity of the user.

In this Review, we introduce the variety of haptic stimuli involved in physical natural interaction and discuss how they can be mapped to different actuation modalities and haptic rendering principles. Although the plurality of tactile stimuli cannot be simulated through a single device, rendering principles can take advantage of those haptic features humans are most sensitive to, resulting in feasible device design and informative haptic feedback. A dedicated section on 'Emerging actuators and device designs' will cover emerging actuating principles that are promising or have been preliminarily tested for wearable haptic applications. We then survey the proposed wearable haptic prototypes, classified on the basis of the body segment these devices are applied to, such as fingertips, hands and arms. Furthermore, we present recent experimental applications of wearable haptic devices in VR and discuss their innovative applications in other fields, ranging from postural correction and navigation assistance to communication and remote control of humanoid robots. Additionally, despite the richness of research and applications in wearable haptics, we highlight relevant challenges still present in the field. These are related to certain fundamental haptic features, not met by technological solutions (that is, spatially distributed cues, such as edge rendering, and the stick–slip condition), and to usability and wearability limits, as emerged by the discussed applications.

Sensory principles for haptic rendering

The creation of haptic sensations relies on eliciting mechanoreceptors that emulate the stimuli encountered during interactions with real physical object surfaces. This recreation is fundamental for evoking a sense of richness in haptics, as it involves the activation of four main types of mechanoreceptors. These receptors — slowly adaptive Merkel (SA-I) and Ruffini (SA-II), and fast adaptive Meissner (FA-I) and Pacinian (FA-II) — differ in their response characteristics concerning adaptation rates, receptor field sizes, and temporal and spatial sensitivities (Fig. 2). Merkel and Ruffini receptors exhibit slow adaptation rates, providing sustained information about surface textures and features, whereas Meissner and Pacinian receptors offer faster responses, albeit with Meissner receptors being more sensitive to smaller details due to their smaller receptor field sizes and Pacinian receptors being attuned to larger-scale stimuli[1].

The haptic perception of an object encompasses various identifiable features. Surface texture, primarily assessed through temporal responses during haptic exploration, contributes substantially. Weight perception is reliant on the kinaesthetic function, whereas compliance pertains to the surface's deformation under force. Moreover, geometrical properties such as orientation, curvature and thermal properties play essential roles in haptic recognition.

Additionally, an array of illusions can be induced, expanding the understanding of haptic sensations. Illusions such as the grill illusion, which associates pain sensations with combined cold and heat stimuli, and the skin stretch illusion that explicates weight perceptions related to size and material or localized skin indentation generating the illusion

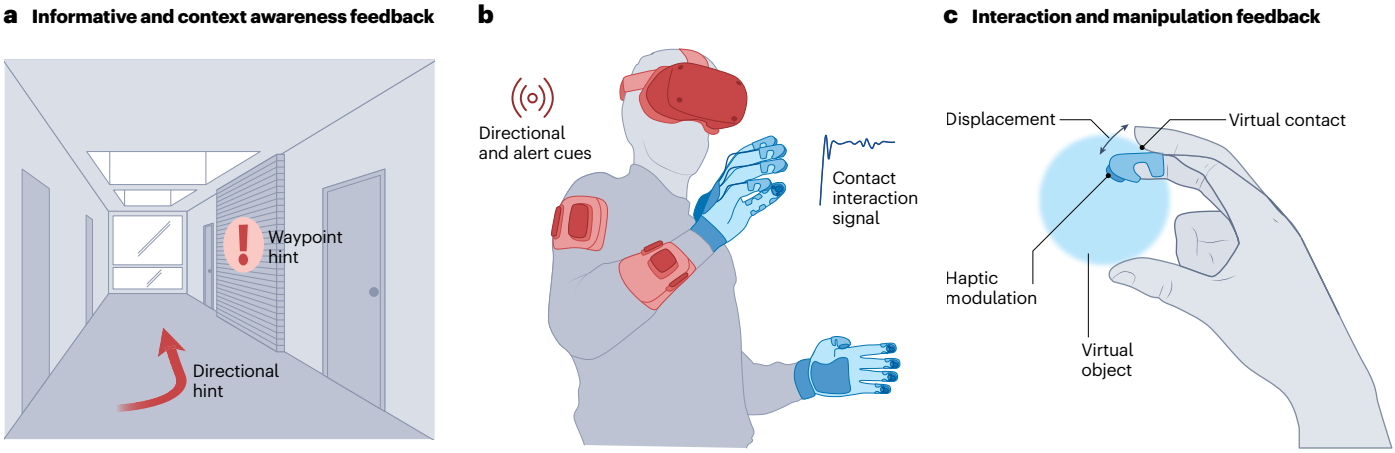


Fig. 1 | Application scenarios of wearable haptics. a–c, Overview of the representative application scenarios of wearable haptics: **a**, for simulating the sense of touch in virtual reality, with stimuli modulated by the collision between the hands and the virtual environment (panel **c**); **c**, for conveying high-level, informative cues supporting the user (panel **b**) in simulated physical tasks.

of edges or holes[2] extend beyond the scope of this Review, but can be exploited to amplify or to mediate perception of features in VR.

The exploitation of these physical principles characterizes the multifaceted approach required to reproduce artificial touch cues that can replicate and simulate the intricate sensory experiences encountered during interactions in the physical world.

Stimulus modalities in VR interaction

Based on the properties of the four types of haptic receptors, several mechanical and multimodal stimulus modalities can be adopted to induce haptic sensations resembling the physical world within a VR environment. The conceptual mapping illustrated in Fig. 3 shows how stimulus modalities can be combined into different actuation principles to elicit tactile sensations at hands.

In VR, haptic stimuli must be coherently mapped to the interaction of the user in the simulated physical world. It is useful at this point to underline the existence of three different entities: the physical user body, the visualized virtual environment and the physical simulation. Although in the real world these three entities match, in the virtual simulation these can differ, owing to physical or technological limitations, or owing to intentionally induced illusions.

Clearly, the virtual world cannot impose physical constraints on the real user, although wearable devices are used to provide the sense of touch. For example, if the user places a hand on a virtual table, wearable haptic devices can generate some tactile perception at the collision, but by no means can apply overt kinematic forces impeding the physical hand to compenetrate the virtual table. The duality between perception of kinaesthetic forces at the level of body kinematics and of tactile perception through skin mechanoreceptors has been already formalized in the literature by the sensory subtraction concept[3].

Various contact interactions are therefore possible, depending on the complexity of the physical simulation in modelling the virtual hand–object collision[4]. These can involve distributed contact points, usually simplified with lumped interaction points to better match physical simulation and rendering capabilities.

The following points summarize conventional features and stimulus modalities adopted in VR rendering.

Contact transition and normal force. Rendering of the transition from the no contact to the contact condition is important both for realism and for informative feedback, regarding also properties of the virtual object such as its stiffness[5]. A recent study has shown how augmentation of transient vibrations, together with visuo-haptic illusions in VR, can artificially modulate the perceived object stiffness[6]. Contact transient rendering is also suitable for rapid interactions, such as with a keyboard, as explored in immersive VR[7].

Once the virtual finger is in contact, a conventional rendering strategy consists of rendering the force opposing the virtual collision, hence in the normal direction of the finger pad. Conventionally, a linear function can be set between the indentation of the finger through the virtual contact surface and the output force applied by the haptic device to the user's finger pad[8].

Lateral stretch and the stick–slip condition. Although normal force rendering might be thought the most immediate and intuitive interaction modality to be rendered in a virtual contact, lateral forces prove informative too. For example, considering the aforementioned sensory substitution principle, the weight of an object can be rendered by tactile feedback only, and in particular through lateral forces applied to finger pads. Different approaches and wearable haptic designs have investigated rendering of such features in VR[9–11].

Closely related to the rendering of lateral forces is the stick–slip condition. This is relevant in VR interaction as well in other wearable haptic applications, such as teleoperation and prosthetics. Already in simple manipulation tasks, such as pick and place, healthy individuals accurately and constantly modulate grasping forces, naturally holding grasping just above the level of static friction required to avoid slippage[12,13]. In VR, as the feedback modalities are limited, the safety margin in terms of forces required to grasp and lift an object is increased[14].

Although extremely relevant in manipulation, rendering of the stick–slip condition is challenging. This results in various combined haptic stimuli, starting from slow modulation of normal indentation and lateral stretch in the stick condition, then abruptly modulating into a decrease of the lateral stretch intensity, a decrease of the area of contact, and introduction of noise and vibrations during the slip condition.

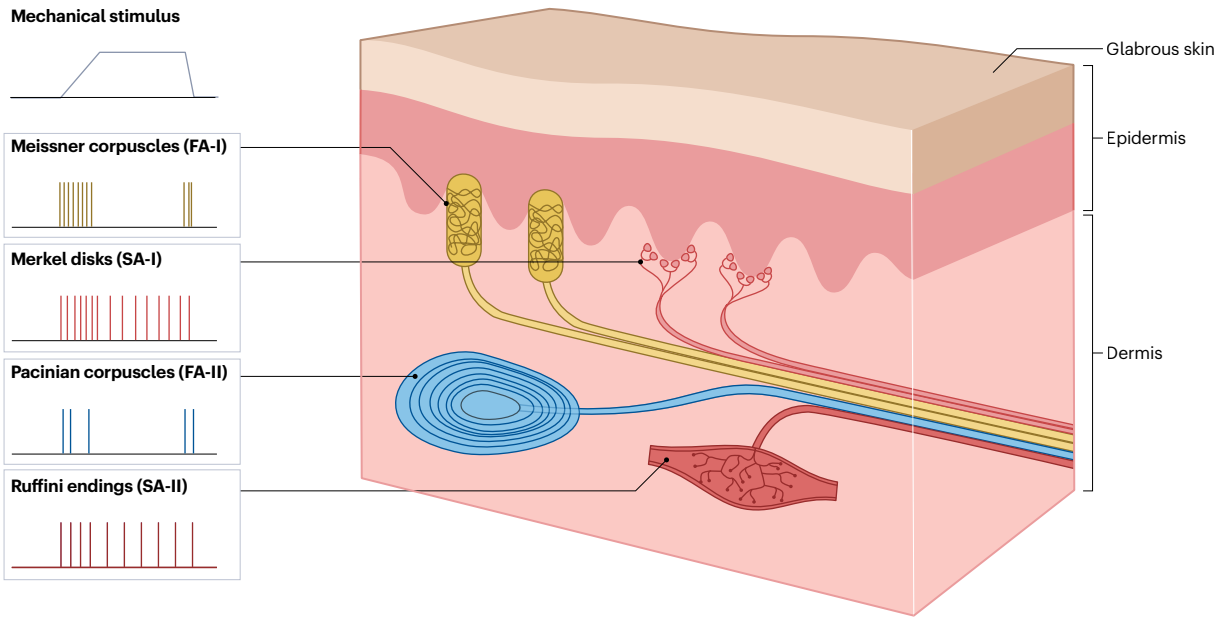


Fig. 2 | Typologies of mechanoreceptors. The four types of mechanoreceptors in finger-pad tissues, and their location in the epidermis and dermis. Graphs show the typical firing rate of each typology in response to a reference mechanical stimulus. During dynamic transients, an intense firing rate is noticeable for all of the typologies, whereas differences between slowly adaptive (SA) and fast adaptive (FA) mechanoreceptors emerge in the steady condition of the reference stimulus.

Vibration and texture rendering. Vibrotactile feedback consists of the dynamic modulation of pressure to skin mechanoreceptors. Rapidly adaptive mechanoreceptors — Pacinian and Messner corpuscles — are highly sensitive to dynamic pressure waves, in a frequency band that extends to audio perception[15]. It has been shown that vibratory-only feedback can resemble perception of real recorded textures, allowing to render artificial textures on otherwise smooth surfaces, put in vibration by the same recorded signals[16]. In addition, methods to synthesize virtual textures in real time, rather than using pre-recorded signals, have been proposed for tool–surface interactions and handheld devices equipped with wide bandwidth linear actuators[17]. These synthesis models enable to render textures based on multiple modulation signals, such as sliding velocity and pressure. In immersive VR, similar approaches can be adopted to modulate wide bandwidth signals within fingertip devices[18,19]. However, to fully resemble the experience of a physical texture exploration, the rendering of the relative sliding between the finger and the virtual surface must be addressed, similarly to the above-described stick–slip rendering.

Surface orientation. Regarding surface exploration and orientation of the contact surfaces, haptic devices and experiments aimed at rendering of virtual shapes with mixed kinaesthetic and tactile feedback of surface orientation were first presented[20,21]. Experimental findings[22] showed that surface orientation dominates haptic curvature discrimination, supporting development of tactile-only devices to render surface orientation. Fully wearable haptic devices implementing orientation rendering have been proposed[18,23]; however, it is a challenging task to lump surface orientation at one contact point, considering both the spatial distribution of the contact area in the virtual world and the physical shape of the end effector of a fingertip haptic device — a circular plate[24].

Thermal feedback. Beyond absolute temperature rendering, thermal feedback can enable discrimination of virtual materials. According to their thermal conductivity, different heat flow and temperature transients are generated at the contact transition with the finger. Wearable devices have been proposed and applied to VR settings with the aim of simulating contact with different materials[25,26]. Beyond simulation, a thermal illusion called the 'grill' illusion[27,28] can be exploited to amplify perception of heat. This is based on thermal gradients that can be artificially generated in haptic devices by nearby plates with different controlled temperatures.

Electrotactile feedback. Electric stimulation of tactile mechanoreceptors, although not a method of natural tactile interaction, can be used to provide tactile sensations far below the pain threshold. This technique is particularly valuable in wearable haptic devices owing to its potential for fine spatial resolution and lightweight rendering devices. However, the high variability of the perceived feedback, depending on different skin conditions, limits the ability to obtain consistent rendering.

Pseudo-haptics and delocalized feedback. Whereas interaction in VR comprises fingertips and hands, other body segments can be considered for haptic rendering through wearable devices. The pseudo-haptics approach aims at taking advantage of certain haptic illusions generated when combined with congruent modulation of the visual feedback[29].

Interactions occurring at the hands and fingertips can be mediated in VR by illusory visual feedback (related to the shape and motion of the manipulated objects) to delocalized haptic feedback at the hand or wrist, in various envisaged object manipulation paradigms[30]. Such delocalized feedback — from the finger pad to the finger proximal

phalanxes — can be also exploited to alter perception of stiffness, friction and shape of the objects within a VR environment[31].

Haptic feedback provided at body segments other than the hand and fingertip is also largely exploited in applications going beyond VR, such as prosthetics, or in guidance and navigation scenarios, where the feedback assumes a purely informative role rather than simulating contact interactions.

Actuation principles

Different haptic stimuli modalities can be addressed by specific actuation principles, convenient for mechanical implementation in wearable haptic devices. For example, surface roll can be rendered by a single rigid plate, oriented around the finger pad, in turn implemented by means of serial or parallel kinematic structures. Additionally, lateral force can be rendered by different actuation principles, such as the aforementioned single rigid plate, or through a soft actuated belt surrounding the finger pad. Several of the above-discussed stimulus modalities articulate in multiple geometrical degrees of freedom (DOFs) or can be extended at multiple contact locations. The technological challenge is then in the complexity of the final mechanism design, which can rapidly increase when including multiple stimulation modalities. The final design can easily reach manufacturing limits, considering also the strict dimensional constraints required to ensure wearability of these devices at the fingertips. The size of the actuators and the intensity of the feedback are, consequently, other relevant trade-offs in the design of such miniature devices.

Explored design solutions can implement a single, rigid contact plate, attached to one or multiple DOFs: linear movement, to display normal or tangential forces; and rotations, to render different curvatures or to engage different contact points around the finger pad. Depending on the bandwidth of the actuator and of the mechanical transmission, an actuated plate can also render vibrations and contact transients.

Another interesting method, exploited in particular for arm-worn devices, involves actuated belts; this can provide distributed normal forces (or clenching), lateral forces and, in certain designs, continuous lateral slip. The latter is a challenging stimulus modality to be rendered in wearable devices as an arbitrary long sliding interface would be needed between the skin and the worn device.

Vibrotactile feedback has been by far the most employed and robust way of delivering touch feedback through wearable and handheld devices, including smartphones. It can be generated by eccentric mass or resonant vibration motors, with fully enclosed moving parts and no need for calibrated contact interfaces. Additionally, owing to the sensitivity of mechanoreceptors to certain vibration frequency ranges, low-power, miniature actuators can produce vivid haptic perception.

However, although vibrotactile feedback turns in robust and effective devices, it appears to be limited in the informative content these haptic signals can deliver to the user. Indeed, inertial vibration motors are limited to single-frequency or narrow bandwidth operation, and with dynamically limited modulation features. In analogy with the audio world, vibrotactile feedback resembles a clear and loud beeping acoustic signal compared with the richness of a music track. It is in this direction that novel devices look for modulation of the haptic signal and quality of the interface for transmission to the finger-pad tissues. Modulated vibrotactile signals can be used to render texture and vibrations, and to modulate contact transients as described in the previous section.

Whereas electromagnetic actuators and voice coils have been largely proposed in the past, soft interfaces and pneumatic actuators

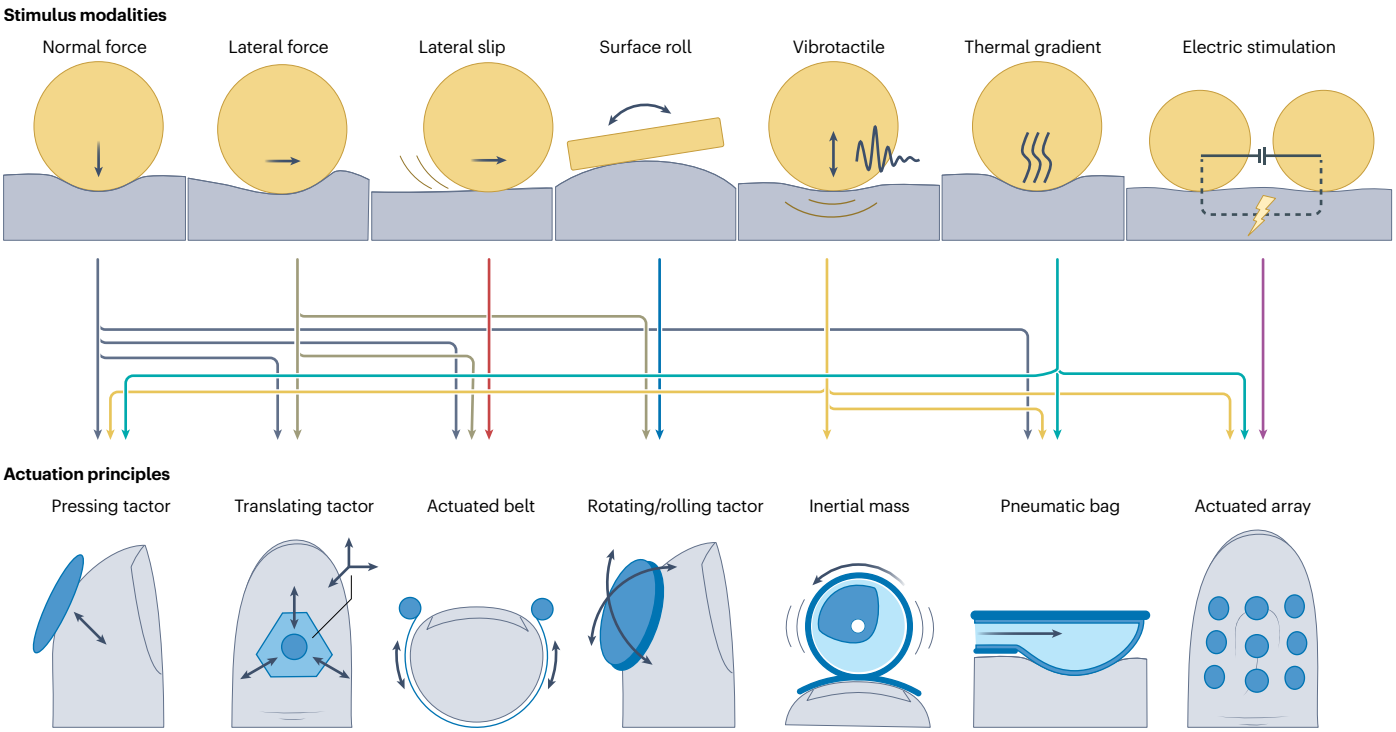


Fig. 3 | Connections between stimulus modalities and actuation principles. A landscape of tactile stimulus modalities (top) and of the different actuation principles (bottom) suitable to convey the given stimuli. Lines show typical implementation solutions: each actuation principle can, in general, be designed to render different stimulus modalities.

are being explored with renewed interest. More advanced prototyping technologies relying on precision 3D printing with soft resins allow more complex soft shapes with embedded inflatable pouches. These allow for soft and compliant adaptation to the finger pad with improved wearability and noiseless transmission of the haptic signal to the skin. The drawback remains the tethering of these devices to remote pneumatic or hydraulic circuits, reducing wearability and limiting the achievable bandwidth in rendering.

Emerging actuators and device designs

The forefront of wearable haptics research has been characterized by advancements in actuation mechanisms, fostering development of novel design solutions. Research in this direction can substantially change the current rendering limits of wearable haptic devices, considering that wearability imposes strict design constraints. The compromise is usually between encumbrance of the device and the quality, intensity and complexity of the actuated haptic features. This section presents the recent actuating methods proposed for wearable haptic design, and then covers both recent and more consolidated device design developed for fingertips, hands and arm body segments.

Sustained interest emerges from the revised literature in developing soft and stretchable haptic actuators, striving to increase wearability and comfort for the user. Emerging actuating methods involve soft and bendable resistive force actuators for haptic gloves, and instability-driven soft pads for vivid vibrotactile feedback in textile and wearable applications. A comprehensive list of emerging actuating methodologies for wearable haptic devices is presented in Supplementary Table 1 and discussed hereafter.

Soft actuated interfaces are convenient for wearable haptics for two main reasons. The soft, compliant surfaces improve contact distribution with the skin, enhancing signal transmission, such as normal force and vibrotactile stimuli. Moreover, the actuated elements can result completely soft and with low aspect ratio, hence improving wearability of the actuated parts. On the other hand, conventional actuating methods for soft interfaces rely on pneumatic or hydraulic systems with tubing and remote pumps, increasing the complexity of the system and limiting its overall wearability. Research advances are mainly focused on alternative methods to actuate soft interfaces, with capabilities of miniaturization and untethered operation.

Liquid-to-gas phase change and thermal expansion in gels[32,33] have been proposed for obtaining untethered, soft deformable pads. These rely on local, electrically driven heating elements, producing a change in volume of a liquid or gel-filled pouch. The method results in compact actuators, but with dynamic limits imposed by heat dissipation capabilities.

Electrohydraulic actuation, which has been widely explored in the field of soft-robotics actuators, was proposed recently for haptic applications, as shown elsewhere[34–37]. Its actuating principle is based on the electrostatic zipping effect: hydraulic pouches, formed by conductive flexible walls and isolated by a thin dielectric layer, function similar to soft capacitors. Under an applied voltage, the electrostatic force causes the walls to zip together, reducing the volume of the hydraulic pouch and inflating the contained fluid towards a soft membrane. In design solutions developed for haptic applications, these pouches can be arranged in arrays of soft, active pads or miniature dots[36], providing vibrotactile feedback to the user through a soft and flexible interface. Additionally, the electro-adhesion effect is explored for modulating friction between sliding layers in resistive flexible actuators. As mentioned before, such actuators are designed to adjust resistive bending forces in haptic glove devices[38,39]. In such actuator typology, adhesion between layers and the modulation of bending resistance are also explored through pneumatic actuation and various interlayer structures, such as teeth-clutching surfaces or grained materials[40,41].

Electroactive elastomers are not fresh to the field; their potential is in the simplicity of operation and of the geometrical structure of actuators, allowing for miniaturization in wearable devices. Under applied voltage, elastomers exhibit deformation that can be directly exploited to provide indentation, pressure or vibrations to the skin. However, the drawbacks are the limited output deformation compared with the input voltage, and reliability. Recent investigations[42–44] have improved the output features, getting closer to implementation of a soft, actuated haptic skin.

Conventional pneumatic actuation for soft interfaces is also further developed in the literature. A soft, instability-driven textile actuator, where arrays of textile pneumatic pads can be mechanically programmed to produce transients and patterns of stimuli, has been developed[45,46]. This method is capable of modulating vibrotactile feedback through a completely soft and thin mechanism, without the use of rigid valves. Another explored method is based on highly deformable pneumatic structures for hand-palm haptic devices[47]. Here, the aim is to provide an overt and more perceivable geometrical change of the whole grasped surface, rather than localized normal forces.

As regards miniature arrays of actuators, various principles are being explored – from pneumatic suction arrays to obtain vivid contact perception, controllable at precise areas of the finger pad, yet with tethered devices limiting wearability[48,49], up to a dense array of vibrating pins, with actuation induced by thermal expansion and relatively high vibrating frequencies[50] or soft, miniature vibrotactile patterns based on the above-described electrohydraulic effect[36].

Fingertip devices

Pioneering developments in fingertip wearable haptic devices have emphasized user perception and interaction. From the comparison between soft pneumatic and vibrotactile feedback systems to the design of soft monolithic devices with embedded actuation, these advancements have delved into diverse actuating principles such as pneumatic, motor-driven mechanisms and electro-adhesive arrays. Rendering features encompass continuous skin slip, mixed pressure/shear and distributed pulses, aiming to simulate varied tactile experiences on the fingertip with precision. A list of the contributions regarding the design of novel wearable haptic devices for the fingertip is summarized in Supplementary Table 2.

Owing to limited size and mass constraints, different design guidelines can be traced. One approach aims at increasing the number of DOFs of movement of a single pad in contact with the finger. These include, beyond the trivial normal force, forces in the longitudinal and lateral directions, and multi-DOF rotation of the pad. The latter feature allows for changing the contact point around the fingertip, also evoking perception of surface orientation. These features are typically implemented through a parallel kinematic structure, typically with gear-motor actuators mounted at the finger dorsum, and links to the moving pad surrounding the fingertip[11,23,51]. Whereas former work exploited rigid parallel kinematics structures (Fig. 4a), recent work explores monolithic, foldable parallel structures actuated by pneumatic or piezoelectric actuators[52,53] (Fig. 4b) or by gear motors[54,55].

Other works aim at extending the rendered haptic modalities of the device, adding to a limited set of motion DOFs of the contact

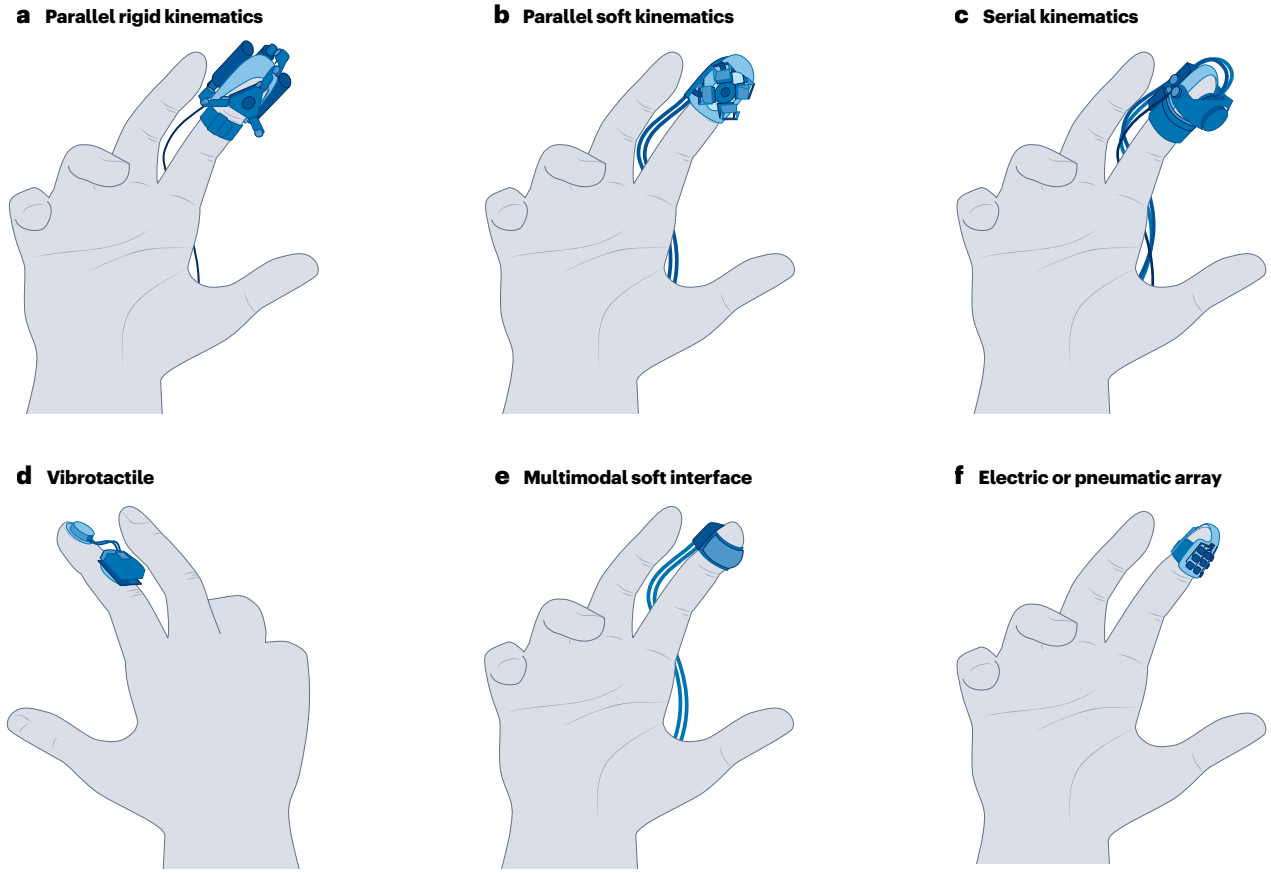


Fig. 4 | Typologies of fingertip haptic devices. Different typologies of fingertip haptic devices. **a**, Based on parallel rigid kinematics and multiple degrees of freedom (DOFs), usually implementing miniature gear motors. **b**, Based on parallel soft kinematics and multiple DOFs, implementing pneumatic or foldable piezoelectric actuators. **c**, Based on serial rigid kinematics and multiple DOFs, the pad in contact with the fingertip can provide different stimuli modalities (vibrotactile, lateral slip, thermal). **d**, Vibrotactile devices, implementing a minimal structure with embedded electronics. **e**, Based on a multimodal soft hydraulic or pneumatic interface, providing modulated normal force and thermal stimuli. **f**, Based on an array of actuators, electrodes or miniature pneumatic bags providing electrotactile stimuli or vibrotactile stimuli, respectively.

plate, features such as continuous lateral slip[56], softness[57] and thermal feedback[26,58] (Fig. 4c).

On the other hand, multiple DOFs imply larger actuation units and mechanisms, possibly limiting wearability of the device and dexterity of the user. A different design approach aims at enhancing compactness and wearability, although with limited rendering features, resulting in extremely compact devices[59] (Fig. 4d). These typically render vibrotactile feedback, intended either as pure vibratory signals generated by inertial motors[60] or as wide bandwidth modulation of the normal force[61–63] generated by a linear voice coil actuator.

Pneumatic and hydraulic devices are also explored, with the benefit of a compliant and soft interface with the skin and limited encumbrance at the finger pad (Fig. 4e). Whereas pumps or liquid reservoirs are typically implemented and tethered to the hand[53,58,64], innovative approaches propose embedded actuators at the fingertip[63,65]. Soft active interfaces are also obtained by means of an actuated fabric band, pulled or clenched around the finger[9,66,67].

The above design and rendering capabilities are lumped to a single actuated pad. Conversely, the dense spatial distribution of mechanoreceptors at the finger pads can be addressed by an arrayed layout of actuators (Fig. 4f). These can be effectively implemented through electrotactile stimulation, with arrays featuring flat and dense distribution of electrodes[68].

Hand devices

Expansive contributions to hand-palm devices have focused on extending haptic sensations to broader areas. Innovations such as the inflatable pop-up interface for palm sensations and 3D shape displays through multi-pouch inflatable handles underscore efforts to create immersive experiences by engaging a larger tactile surface. Modulating inflating pressure and multi-DOF manipulations form the crux of rendering features, promising a comprehensive touch sensation across the hand palm. The main design solutions regarding hand-worn haptic devices are reported in Supplementary Table 3, together with the main haptic glove technologies available as commercial products. We describe hereafter the main design principles adopted by the referenced devices.

Electrotactile and vibrotactile modalities, due to the compactness of electrodes and actuators, have been proposed in the shape of gloves covering a higher number of contact locations with respect to the finger

pads only[69–71], or integrating an additional stimulus modality[10,38,72] – such as Nova 2 and Teslaglove.

Extending haptic feedback to the hand can involve not only tactile perception at the palm but also kinaesthetic feedback related to opening and closing of the hand. To clarify, whereas haptic fingertip devices can produce a localized perception of pressure but no relative force between fingers (hence defined as cutaneous or tactile force feedback), kinaesthetic force feedback results in an overt force, opposing the muscular activation of the user during finger closing. Although haptic gloves could be conceived as active hand exoskeletons, there are substantial technological differences between them. Haptic gloves do not have to assist movements; rather, they must render physical constraints, such as a virtual grasped object. Therefore, actuators can be conceived in the form of compact and strong modulated brakes, opposing the active hand-closing movement performed by the user. This is the main design approach, nowadays consolidated in a series of commercial products based on resistive force feedback such as CyberGrasp, Haptx, Nova 2 and Teslaglove. The structure of the glove is typically based on soft cable transmission[73,74], routed from the hand dorsum or forearm to the finger dorsum as depicted in Fig. 5a, or on rigid links[75,76] – such as Teslaglove.

A more innovative, yet experimental, approach relies on soft actuated bands capable of modulating friction forces in response to bending or elongation. These soft, resistive actuators are mainly based on a pneumatic jamming effect or on electrostatic forces, and have been recently proposed in several haptic glove devices[38–41,77] (Fig. 5b).

Solutions providing contact perception at the broader area of the hand palm have been proposed in the form of parallel kinematic structures[72,78]. These devices use design concepts similar to multi-DOF fingertip devices, with the main body of the device placed at the hand dorsum and parallel actuated arms reaching the hand palm (Fig. 5c).

Arm-worn devices

Advancements in arm-worn haptic devices have targeted specific functionalities and applications. Designs such as wearable skin stretch devices for haptic guidance and force-controlled multimodal haptic bracelets offer tailored solutions for users seeking tactile guidance and interactive experiences on the forearm. Contributions to the design of limb-worn haptic devices are listed in Supplementary Table 4 and discussed hereafter.

Two main stimulus modalities appear as the most consolidated in the literature: clenching actuation (Fig. 6a) and vibrotactile arrays (Fig. 6b,c). Vibrotactile arrays can be implemented in compact and robust devices in the shape of armbands (Fig. 6b), often in combination with other stimulus modalities[79–83], or as actuated matrices[84,85] (Fig. 6c). As regards matrices of vibrotactile actuators, recent investigation on the funnelling effect[86] aims at rendering spatially localized haptic cues with a reduced number of actuators at the edges of the active area.

Clenching actuation, formerly applied at the upper limb[87], presents advantages similar to vibrotactile feedback in terms of robustness, stability and comfortable wearing of the device. The method has been further proposed in more recent devices[79–82,88–90] (Fig. 6a) combined with vibrotactile feedback. As already exploited for the fingertip application, by actuating the fabric band with two motors it is also possible to induce a lateral stretch stimulus. An alternative modality to convey localized skin stretch is represented by a rotating wheel in direct contact with the skin. This method can be implemented in simple and compact design solutions[91–93].

Alternative actuation and stimulus modalities are represented by pneumatic actuators, rendering normal force and vibrations with arrays of four to eight active locations[94–96], pneumatic multi-DOF actuators[97,98] featuring rendering of lateral stretch, and thermal cells[83,99–102].

Applications in VR, AR and telepresence

Application of wearable haptics is continuously evolving and nowadays covers a series of different application areas. The rendering of haptic interactions in VR is surely one of these applications, with most of the aforementioned commercial gloves targeted to this purpose. This is fostered today by the presence on the market of highly 3D visor products, embedding in a consumer-grade product high-quality visual feedback and embedded head and hand tracking. The latter allows for immersion of the user at the level of movements and also natural hand interaction. Supplementary Table 5 summarizes the main experimental applications of wearable haptic devices in VR, AR and telepresence, discussing hereafter the emerged trends.

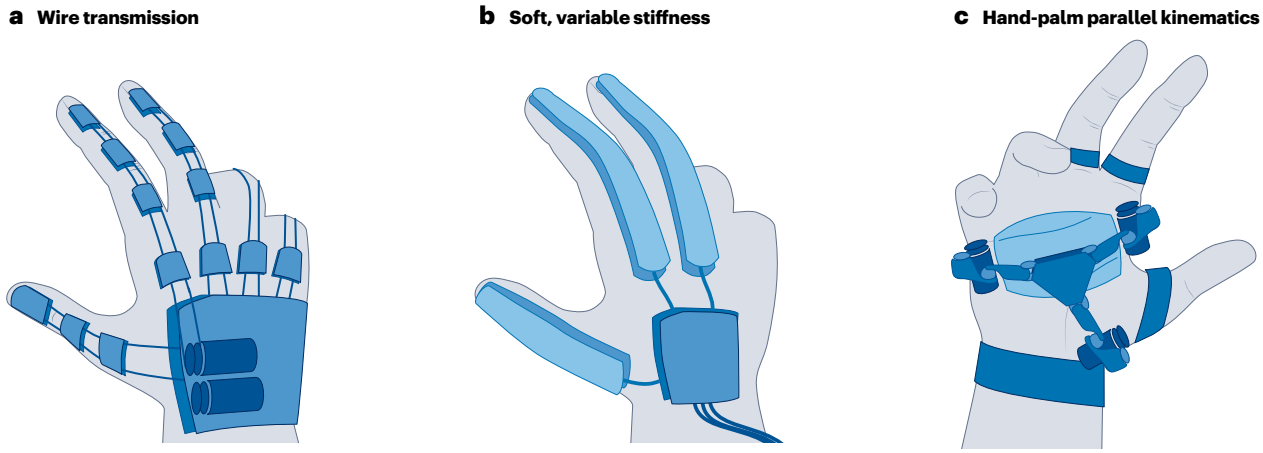


Fig. 5 | Typologies of hand-worn haptic devices. Representative typologies of hand-worn haptic devices. **a**, Haptic gloves based on tendons or rigid kinematic links located at the finger dorsum, actuated using gear motors or electromagnetic brakes. **b**, Haptic gloves implementing soft, variable stiffness bending actuators located at the finger dorsum, based on pneumatic actuation or electro-adhesion. **c**, Based on parallel rigid kinematics located at the hand palm, implementing multiple degrees of freedom (DOFs) providing different contact locations, normal and lateral forces.

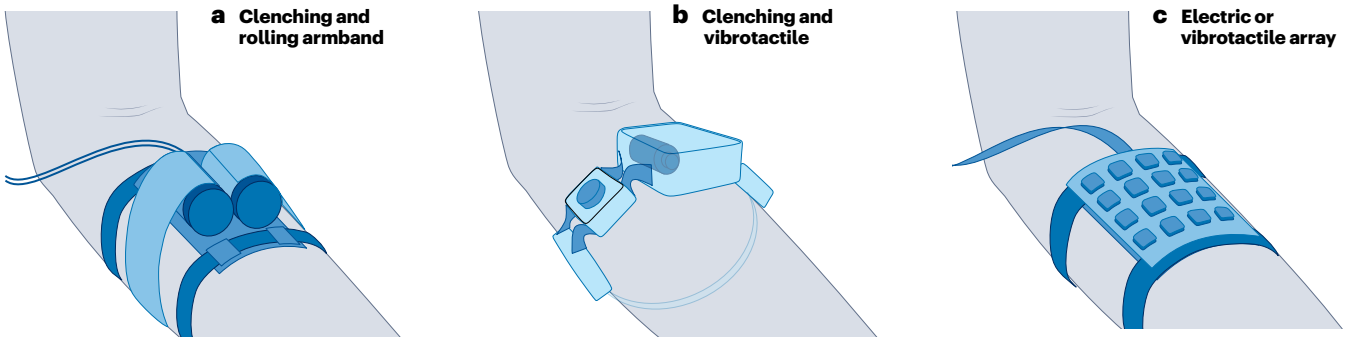


Fig. 6 | Typologies of arm-worn haptic devices. Different typologies of arm-worn haptic devices. **a**, Based on a soft belt actuated by two motors, providing normal (clenching) and lateral forces. **b**, Featuring distributed vibrotactile actuators and a single actuated wire providing normal force (clenching). **c**, Based on a distributed array of actuators, electrodes or miniature vibration motors, providing electrotactile or vibrotactile stimuli, respectively.

In the research field, the use of haptic feedback in immersive VR is being experimented with gamified exercises for neurorehabilitation of children with motor disorders[103] (see Fig. 7). Here, the immersive and gamified set-up can improve engagement of the patient, and therefore their commitment in performing game-oriented manipulation exercises[103].

By immersion of the user in a virtual world and its mediated perception[104], applications involving embodiment of a virtual avatar with haptic feedback are also explored for anxiety and pain treatment[105,106]. In this line of innovative applications based on virtual embodiment and sense of touch, several studies use these technologies to simulate crowded environments, investigating critical emotional and behavioural response[107–109]. Virtual immersion and haptic feedback is also explored in the art and cultural field, to enhance the experience of both art pieces and live performances[110,111].

Training in the medical, industrial and service fields is another consolidated application for VR; here, however, the sense of touch and physical interaction appears critical to the point that the so-called 'passive haptics paradigm' seems preferable[106,112,113]. Here, physical mock-ups and user interfaces are matched with the VR visual representation.

Telepresence, although not a VR application, is included here due to the similarity of interaction and feedback to the user in a virtual presence. Recent literature is rich with telepresence systems immersing the user in remote humanoid robots, fostered by international challenges and competitions on the topic[114]. The sense of touch at the hands is enabled through wearable haptic interfaces, coupled or not with additional robotic interfaces for kinaesthetic feedback at arm level[115–118].

Applications beyond VR

Wearable haptics finds an increasing number of applications beyond VR. In these fields, the rendering location usually shifts from the hand to the limbs and torso. This shift is also due to the nature of the rendered signals, which are more related to high-level information and cues rather than the simulation of physical interaction. The main contributions representative of applications of wearable haptic devices in different fields are outlined in Supplementary Table 6. In the following paragraph, we discuss in detail the emerging application areas.

The use of wearable haptics for smart prosthetics and wearable robotics at large is today a rich application area. In the case of amputees, the sensory feedback from the prosthetic hand interaction can be provided at a remote location of the body, such as the arm or forearm, improving grasp control[119]. Recent studies evolve the methods based on different stimulus modalities and devices, such as clenching[120–122], dense vibrotactile arrays[84] and contact location[123]. In a previous study[124], soft haptic actuators were integrated into the prosthetic socket itself, easing usability of the system and providing modulated pressure feedback to the user.

Also in the medical field, tactile augmentation at finger pads is being experimented for sensorimotor rehabilitation[125], with vibrotactile feedback[126] used for 'nudging' the user, promoting activity of the impaired limb during stroke rehabilitation.

Additionally, the application of wearable haptics in language and communication is a rich and active research area[127]. Recent literature has reported on the exploration of effective encoding and rendering methods to produce distinguishable stimulation patterns for the user[93,128,129].

This field is closely related to the use of wearable haptics for guidance and situation awareness, where communication of high-level information and alerts to the user is conveyed through the haptic channel[80,84,130]. A central topic involves guidance in navigation, especially for visually impaired individuals[131,132]. Yet recent studies extend the concept of directional guidance to posture and ergonomics, using wearable haptic devices to hint to the user postural corrections during loading tasks[89,133].

Emerging application fields involve the use of the haptic sensory channel to assist mental states, such as in anxiety treatment[134] and stressful tasks[135] by means of haptic vests with extended pneumatic pouches applying distributed pressure stimuli to the body.

Beyond teleoperation, wearable haptics can be used to enable a high-level communication channel between the robot and the operator, to convey interaction cues, guidance and collision awareness alerts[136]. Along with the widespread field of collaborative robotics, 'wearable' haptic devices are also proposed to be worn on the robot side[137], to communicate with the human grasping the robot during manual teaching operations.

Related to this, we are currently assisting a major revolution in humanoid robotics, with the development of advanced robotic platforms aimed at replacing human workers in advanced manufacturing and industrial tasks, requiring touch-enabled advanced human–machine interfaces both for human–robot collaborative assembly and for remote telepresence in the execution of dexterous manipulation tasks. The wide spread of this technology, with both the proposal of

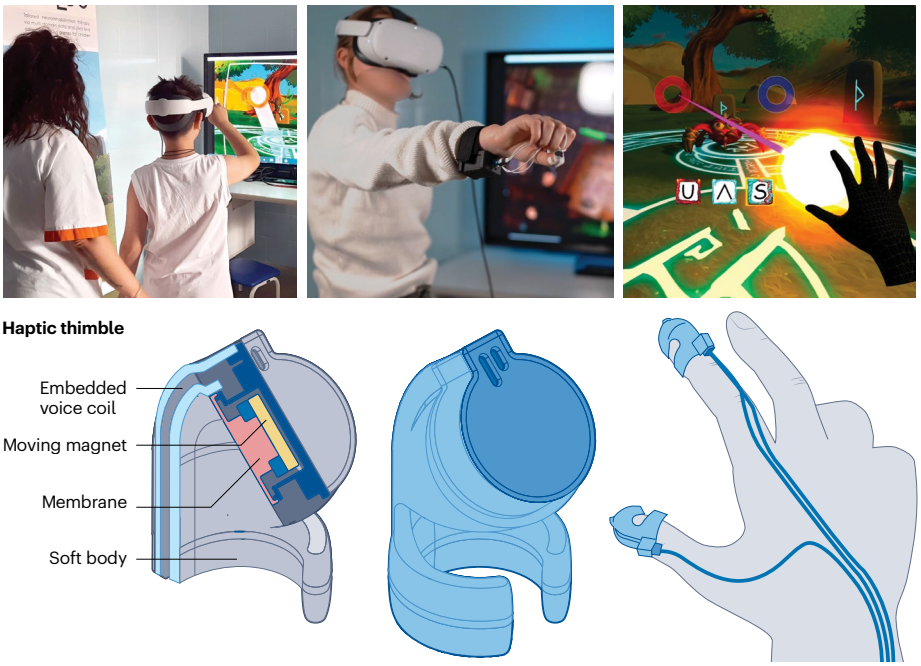


Fig. 7 | Application of wearable haptics for neurorehabilitation in VR. Representative application of wearable haptics for rehabilitation of children with cerebral palsy[103]. Motor exercises are provided as serious games in immersive virtual reality (VR) and with hand interaction. The child, supervised by the therapist, performs pick and place, prono-supination and trajectory tracking tasks (top). Haptic feedback aims at higher immersion, engagement and more natural sensory feedback in such finalized motor tasks. The implemented haptic thimbles (bottom) have been developed taking advantage of soft materials (the thimble body and the actuated membrane), to obtain compactness, compliance with different hand sizes and compatibility with vision-based hand tracking.

low-cost whole-body teleoperation[138] and the advent of robotic avatars for telepresence — a remotely controlled robot capable of representing a person in a location where they are not physically present[114], with manipulation capabilities and touch sensations for the operator — opens a relevant field of application of whole-body and hand-based wearable haptics, posing challenges and perspectives for technological development aimed at extreme wearability and high haptic fidelity.

Conclusions and outlook

The convergence of research studies on wearable haptic devices across different body placements highlights a paradigm shift in human–computer interaction. The use of diverse actuating principles — ranging from pneumatic to motor-driven mechanisms — and stimulus modalities outlines the need for enhancing immersive experiences in XR. The analysis of the innovations presented in these studies collectively emphasizes the potential of wearable haptic devices to revolutionize sensory interactions and redefine human perception in virtual and augmented environments. At the same time, there are still unmet needs emerging from the directions of recent research in device design and from the applications.

When considering dimensions and usability of wearable devices, VR visors have seen tremendous technological advances in recent years, going from heavy head-mounted displays with limited resolution — such as NVIS SX111, largely used in VR research set-ups — to high-performance, lightweight consumer-grade products, also featuring wireless operation and embedded, markerless head and hand tracking.

Although similar progress could be expected for wearable devices, only a few design approaches have led to considerably thin and lightweight devices, and this together with a trade-off with respect to the haptic rendering effective capabilities. Approaches based on vibrotactile signals result, usually, in smaller devices, due to the advantage of the high sensitivity of fast-adapting mechanoreceptors. On the other hand, with these methods the rendering is limited to vibrations, often with fixed or limited frequency range, not including the variety of haptic stimuli. Electrotactile stimulation is another approach resulting in slim and high-density haptic interfaces, yet with possible large variability in the perceived intensity between different subjects and operating conditions.

We foresee that future research efforts will focus on investigating soft and wearable interfaces, fostering the development of personalized haptic interfaces tailored to fingertip sizes and shapes. Recent emerging actuator typologies, such as miniature foldable actuators and the electrohydraulic actuation principle, can in fact lead to soft, multi-DOF and/or high-resolution wearable interfaces, further amplifying the potential for nuanced and immersive haptic feedback.

The theme of improving wearability and density of actuators also requires novel actuating principles: for instance, the electro-adhesion zipping effect can obtain compact actuators with higher energy density than electromagnetic motors, and these can be directly coupled with hydraulic soft interfaces for transmitting signals to the skin. Investigation of the electro-adhesion principle has emerged also in glove-like devices, obtaining thin, flexible actuators applied as a modulated braking mechanism opposing finger closing. Other research directions can explore alternative strategies for rendering continuous signals using discrete or vibration-based tactile transients, encoding information otherwise carried by continuous force components. The latter are less efficiently rendered with compact actuators[121,139,140].

Spatially distributed haptic feedback also emerged in the form of an array of actuators, applied both at the arm and the finger pad, and through investigation of the funnelling effect, emerging as well as a rendering method to convey haptic sensations with a reduced number of actuators[71,86]. Actuator arrays are reaching higher densities and can be implemented to convey interesting distributed tactile sensations and information[84].

Additionally, those actuators are mostly limited to vibrotactile feedback, owing to the output of the compact actuators. Devices featuring variable shape by means of inflatable multi-pouch structures

also show innovative potential, but still operate at the larger scale of the hand palm, rather than at the finger-pad level. Conversely, investigations into the perception of spatially distributed skin stretch at the finger pad have shown the precious capability of creating illusions of edges[141], whose perception is important in dexterous manipulation, and can foster development of devices using a similar effect.

Considering the aspects of sensory feedback, information on the stick–slip condition is continuously used by humans to modulate grasping in manipulation. However, the combined haptic rendering stimuli (low to high dynamics, normal and lateral strain, noise modulation and/or surface sliding), the fact some features are highly subject-dependent – vibrations depending on the shape of finger-pad ridges – and the fact that humans are so used to perceiving them in any real manipulation task, all make the rendering of the stick–slip condition a challenge and a still unsolved problem in wearable devices. In relation to the slip condition, the relative motion between the finger and a physical surface is at the same time a difficult feature to provide in a wearable fingertip device (which has no relative motion with the finger) and a powerful perception condition in natural exploration[142], amplifying the acuity of surface haptic features. Effort in this direction, that is, using a continuous sliding surface, has emerged only in a few studies, suggesting the underlying technical challenges.

Delocalizing the haptic feedback from fingertips to more proximal segments of the body (phalanxes, hand dorsum or wrist) emerges also as a broadly explored solution that proves compliant with AR and real physical interaction by the user. Here, both vibrotactile and modulated clenching stimuli are the main haptic signals provided, resulting in simple, yet robust and stable, wearable devices, easily adapted to different body segments.

Additionally, the application spectrum of wearable haptics is expanding significantly. Studies focusing on performance analysis for limb postural adjustment, integration of soft haptics into prosthetic sockets, immersive virtual environments for neurorehabilitation and wearable haptics for tactile augmentation highlight the versatility of wearable haptic devices. These applications utilize diverse feedback systems, including distributed vibrotactile, skin stretch, clenching and modulated pressure, offering tailored solutions for various therapeutic and prosthetic purposes. For example, vibrotactile[119] and modulated pressure[124] feedback have been explored to convey contact events and modulation of the grasping force in a prosthetic hand application. Skin stretch has been used to provide informative haptic feedback at the finger pads within manipulation rehabilitation exercises, provided in the form of serious games in VR[103].

Also emerging are the potential and the need for more advanced and realistic interaction in VR. Different recent VR applications, including training in the clinical, industrial and service fields, make use of alternative approaches, such as the passive-haptic paradigm, to obtain the needed level of fidelity in physical interactions. Similarly, certain approaches involving the sense of touch in consumer-oriented applications use the vicarious-haptics and pseudo-haptics paradigms, probably driven by considerations on usability of wearable interfaces.

Emerging design methods in wearable haptics also have the potential to reshape the perceptual experience through approaches augmenting the perceived physical world. The above-mentioned pseudo-haptics approach and delocalized rendering, or fingertip interface design suitable for augmenting interaction with tangible objects, can find promising applications within AR settings, but is as yet a scarcely explored field when associated with immersive, wearable haptic feedback.

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A.F. and D.L. contributed equally to the manuscript. A.F. and D.L. researched data for the article, substantially contributed to discussion of content, wrote the manuscript and reviewed/edited the manuscript before submission.

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